

● HARD

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● ANSWER KEY

Command of Evidence

09

10 answers · Rationale-rich · Teach from doc

Q1**A**

A. above 344 km/s but below 378 km/s.

Choice A is the best answer because it uses data from the table to give the range of velocities for the LMC from the 1980 value (344 km/s) to the 2006 value (378 km/s), thereby effectively completing the text. The text indicates that before 2006, all the estimated velocities of the LMC were within the range necessary to maintain orbit around the Milky Way galaxy. It then indicates that, according to Besla and colleagues, the 2006 estimate of 378 km/s is too high to maintain that orbit. This strongly implies that if the 1980 value (344 km/s) is below the orbital threshold, and if Besla and colleagues are correct that the 2006 value (378 km/s) is above that threshold, the maximum orbital velocity for the LMC must be somewhere in the range from above 344 km/s to below 378 km/s.

Choice B is incorrect. The text indicates that the 2006 velocity estimate for the LMC (378 km/s) was the first estimate that exceeded the velocity needed to maintain orbit around the Milky Way. Thus, the 1980 estimate of 344 km/s and the 1994 estimate of 297 km/s must both be below the maximum possible orbital velocity for the LMC. Choice C is incorrect. The text states that Besla and colleagues' analysis found that the velocity from the 2006 study (378 km/s) was too high for the LMC to maintain orbit around the Milky Way. Therefore, if a velocity of 378 km/s is too high, an even higher velocity will also be too high. Choice D is incorrect. The text indicates that the 2006 velocity estimate for the LMC (378 km/s) was the first to exceed the velocity range required to maintain orbit around the Milky Way. Thus, the 1994 estimate of 297 km/s must be below the maximum possible orbital velocity for the LMC.

Q2

D

- D. At current surface temperatures, more than 80% of the sites cannot use subsurface thermal pollution to meet any portion of local home heating needs, but at the maximum plausible surface temperature, that percentage drops below 20%.
-

Choice D is the best answer. The researchers concluded that as we approach maximum plausible surface temperatures, there will be a larger percentage of sites where thermal pollution could contribute to meeting home heating needs. By showing that only a small percentage of homes can currently use thermal pollution for home heating, and that this percentage would grow much larger at maximum plausible surface temperatures, this choice supports the researchers' conclusion.

Choice A is incorrect. We do not know how many sites could have all (i.e., 100%) of their local heating needs met by thermal pollution, as the graph only classifies sites by whether "0%," "Up to 25%," and "More than 25%" of heating needs could be met. Choice B is incorrect. The graph is not depicting need for supplemental heating from thermal pollution, but rather potential to use thermal pollution for supplemental heating. Choice C is incorrect. The graph indicates that, at current surface temperatures, less than 10% of sites can meet 25% of local home heating needs and that more than 80% of sites cannot meet any local home heating needs.

Q3

D

- D. Cownose rays have increased in regional abundance as dusky sharks have decreased in regional abundance.
-

Choice D is the best answer because it presents a finding that, if true, would most directly support the researchers' hypothesis about the connection between the dusky shark population decline and the eastern oyster population decline. The text indicates that although dusky sharks don't usually eat eastern oysters, they do consume cownose rays, which are the main predators of eastern oysters. An increase in the abundance of cownose rays in the region in response to a decline in the abundance of dusky sharks would directly support the researchers' hypothesis: a higher number of cownose rays would consume more eastern oysters, driving down the oyster population.

Choice A is incorrect because a finding that there's an association between a decline in the regional abundance of some of dusky sharks' prey and the regional abundance of dusky sharks wouldn't directly support the researchers' hypothesis that a decline in dusky sharks has led to a decline in eastern oysters in the region. Although such a finding might help explain why shark abundance has declined, it would reveal nothing about whether the shark decline is related to the oyster decline. Choice B is incorrect because a finding that eastern oyster abundance tends to be greater when dusky sharks and cownose rays are present than when only dusky sharks are present wouldn't support the researchers' hypothesis that a decline in dusky sharks has led to a decline in eastern oysters in the region. The text indicates that the sharks prey on the rays, which are the main predators of the oysters; if oyster abundance is found to be greater when rays are present than when rays are absent, that would suggest that rays aren't keeping oyster abundance down, and thus that a decline in rays' predators, which would be expected to lead to an increase in the abundance of rays, wouldn't bring about a decline in oyster abundance as the researchers hypothesize. Choice C is incorrect because a finding that consumption of eastern oysters by cownose rays increased substantially before dusky sharks declined in regional abundance wouldn't support the researchers' hypothesis that the decline in dusky sharks has led to a decline in eastern oysters in the region. Such a finding would suggest that some factor other than shark abundance led to an increase in rays' consumption of oysters and thus to a decrease in oyster abundance, thereby weakening the researchers' hypothesis.

Q4

D

- D. “Alexandra drew her shawl closer about her and stood leaning against the frame of the mill, looking at the stars which glittered so keenly through the frosty autumn air. She always loved to watch them, to think of their vastness and distance, and of their ordered march. It fortified her to reflect upon the great operations of nature, and when she thought of the law that lay behind them, she felt a sense of personal security.”
-

Choice D is the best answer because it most effectively uses a quotation from *O Pioneers!* to illustrate the claim that Alexandra Bergson takes comfort in understanding the world around her. In the quotation, Alexandra is described as enjoying looking at the stars and feeling a “sense of personal security” when she contemplates nature’s order and its governing laws. This suggests that Alexandra takes comfort in understanding the world around her.

Choice A is incorrect because the quotation expresses how Alexandra Bergson attempts to meet difficult situations with determination, not how she takes comfort in understanding the world around her. Choice B is incorrect because the quotation expresses “how much the country meant to” Alexandra Bergson, not how she takes comfort in understanding the world around her. In detailing some of the wildlife surrounding Alexandra, the quotation conveys that nature is important to her but not necessarily that it gives her comfort. Choice C is incorrect because the quotation describes Alexandra driving her wagon down a highway at night; it doesn’t describe how she takes comfort in understanding the world around her or address how she’s feeling as she drives off.

Q5

C

- C. “Stevens’s ambassadorial sandpaintings are best understood not as self-contained objects but as reminders of the public creations of the sandpaintings, during which Stevens conducted appropriate *ikaah* rituals and encouraged viewers to closely track his movements and subtle shifts in the sand throughout the process.”

Choice C is the best answer because it most directly supports the text’s claim that Stevens strove to convey the ephemeral and dynamic characteristics of *ikaah* as a cultural practice. The text challenges the characterization of Stevens’s preserved sandpaintings as mere “static objects” that cannot “authentically represent” a practice defined in part by the absence of an end product; it claims instead that this “overly object-focused” interpretation overlooks how Stevens preserved the essential characteristics of traditional *ikaah* practice. This quotation supports the claim by explaining that Stevens’s sandpaintings should be understood “not as self-contained objects but as reminders of the public creations of the sandpaintings,” during which Stevens “conducted appropriate *ikaah* rituals and encouraged viewers to closely track his movements and subtle shifts in the sand throughout the process.” The fact that Stevens created the sandpaintings in a public setting and invited viewers to attentively engage with this process supports the claim that he sought to portray *ikaah* as a dynamic practice rather than as a static product. Further, the inherently impermanent nature of such a performance evokes the ephemerality that characterizes *ikaah*.

Choice A is incorrect because it contradicts rather than supports the text’s claim that Stevens strove to convey the ephemeral and dynamic characteristics of *ikaah* as a cultural practice. This quotation argues that Stevens’s preserved sandpaintings “should not be confused with authentic *ikaah*” and emphasizes that authentic *ikaah* “cannot be extricated from a practice that is intentionally and necessarily transitory.” This view aligns with the “overly object-focused” perspective that the text argues against, rather than supporting the claim that Stevens successfully conveyed *ikaah*’s ephemeral and dynamic characteristics. Choice B is incorrect because it focuses on reconciling the tension between ephemeral and permanent forms rather than supporting the claim that Stevens actively conveyed *ikaah*’s ephemeral and dynamic characteristics. While this quotation acknowledges an apparent contradiction, it suggests that Stevens was accommodating US preferences for permanent artworks; it doesn’t demonstrate how he communicated the ephemeral and dynamic nature of traditional *ikaah* practice. Choice D is incorrect. Although the quotation mentions Stevens’s “temporary and performative role” as a gift-giver, it doesn’t address how Stevens conveyed the ephemeral and dynamic characteristics of *ikaah* itself during his ambassadorial activities. The quotation focuses on Stevens’s role in transmitting knowledge and representing cultural dynamism but doesn’t specifically support the claim about his efforts to communicate the essential characteristics of *ikaah* practice. Further,

the quotation incorrectly implies that Stevens shared his knowledge of preservative additives; the text doesn't establish that he revealed how the sandpaintings were preserved.

Q6

C

C. *DptB* may have developed as a specific defense against *A. sicerae*.

Choice C is the best answer because it most logically completes the text based on supporting data in the graph. The text indicates that in the fly *D. melanogaster*, *DptA* and *DptB* are genes that encode peptides that both fight pathogens and promote beneficial microbes. Researchers tested *D. melanogaster*'s resistance to *P. rettgeri* and *A. sicerae* bacteria based on which variation of the peptide-encoding gene the flies exhibit: *DptA* silenced (referred to as type A), *DptB* silenced (type B), or both silenced (type AB). The text also indicates that resistance to *P. rettgeri* correlates with *DptA* activity but not with *DptB* activity (which would manifest as type B flies surviving at a higher rate than other fly types when exposed to *P. rettgeri*). The graph shows the post-*A. sicerae* infection results, which indicate that *DptB* activity was most strongly associated with survival, whereas *DptA* activity was not (manifesting in the graph as the type A flies having greater survival rates than the other fly types). In other words, when *DptA* activity was silenced, the flies showed relatively high survival rates, but when *DptB* activity was silenced, whether on its own or in conjunction with *DptA* activity being silenced, survival rates were low, suggesting that *DptB* may have developed as a specific defense against *A. sicerae*.

Choice A is incorrect. The graph suggests that *DptA* activity is associated with a low rate of survival, not a high one. Furthermore, the graph shows results for flies where *DptA* alone was silenced, *DptB* alone was silenced, and both were silenced and thus does not show any flies with activity in both *DptA* and *DptB*, which would be necessary to determine whether *DptA* conferred defense against *A. sicerae* in the presence of *DptB*. Choice B is incorrect. Only two bacteria species were considered in the text: *P. rettgeri* and *A. sicerae*. The text and graph taken together suggest that activity in *DptA* is associated with resistance to *P. rettgeri* while *DptB* activity is not, and that *DptB* activity is associated with resistance to *A. sicerae* while *DptA* is not. There is no further information to suggest one genetic type confers resistance to a greater number of pathogens than the other. Choice D is incorrect. The graph does not address flies with activity in both *DptA* and *DptB*. All flies represented in the graph had one or both of *DptA* and *DptB* silenced, or inactive.

Q7

D

- D. In Akumal Bay and Tulum, delta 15-N values fluctuated considerably across the three measurements made from October 2016 to October 2017.

Choice D is the best answer because it describes data from the table that support the underlined conclusion. The text accompanying the table states that delta 15-N values (a measure of the ratio of two nitrogen isotopes) can be used as an indirect means of measuring the ratio of human wastewater to water from uncontaminated sources—the higher the delta 15-N values, the greater the presence of human wastewater. The text goes on to mention an experiment conducted by Camacho-Cruz and colleagues measuring delta 15-N values in seagrass on the Yucatan peninsula at several sites, four of which are presented in the table accompanying the text. Two of the sites noted in the table, Akumal Bay and Tulum, had intermediate tourism development, and two others, Mahahual and Xahuayxol, had low tourism development. It is reasonable to infer that sites with intermediate tourism have a greater human presence than those with low tourism and therefore are subject to greater amounts of wastewater and exhibit higher delta 15-N values in seagrass. However, Camacho-Cruz and colleagues found that the increase in delta 15-N values was not constant. The table supports this conclusion because the two intermediate-tourism sites (Akumal Bay and Tulum) had delta 15-N values that fluctuated considerably—both had higher delta 15-N values than the two low-tourism sites (Mahahual and Xahuayxol) in October 2016 and October 2017, but in February 2017 the delta 15-N values for the intermediate-tourism sites dropped significantly (in fact, Mahahual, a low-tourism site, had a higher delta 15-N value for that period). The fluctuations in values in Akumal Bay and Tulum support the conclusion that these sites experienced influxes of human wastewater intermittently.

Choice A is incorrect because it inaccurately describes data from the table: in February 2017 the delta 15-N value in Akumal Bay was higher than the delta 15-N value in Xahuayxol, not lower. Choice B is incorrect because the absence of data for delta 15-N levels in Akumal Bay in February 2016 would neither prove nor disprove that the presence of human wastewater at Akumal Bay and Tulum was not constant. Despite the unavailable data, there is still enough information to conclude that delta 15-N levels at both sites fluctuated significantly and that influxes of human wastewater therefore happened intermittently. Choice C is incorrect because a constancy in delta 15-N values would not indicate that the influx of human wastewater was intermittent, but rather the opposite. Moreover, the two periods selected are not indicative of general trends in the whole table (for example, the values fluctuated significantly in Akumal Bay between February 2017 and October 2017, and in Tulum between October 2016 and February 2017).

Q8

C

- C. didn't consistently demonstrate a significant increase in REM sleep after their period of deprivation in the water.
-

Choice C is the best answer. If REM sleep were homeostatically regulated in fur seals, then all the seals would compensate with REM levels significantly over baseline after going weeks without REM. We'd also expect the seals to maintain those elevated REM levels for some time. Since seals B and C return very quickly to baseline REM levels, this suggests that REM sleep in fur seals may not be regulated homeostatically.

Choice A is incorrect. This doesn't support the conclusion. If REM sleep were homeostatically regulated in fur seals, then we'd suspect the seals to sustain REM levels well above baseline for a prolonged period in order to compensate for weeks of REM deprivation while in the water. Whether or not there's a reduction in REM sleep from day 1 to day 2 doesn't tell us how REM sleep on those days relates to baseline, which is where our focus should be. Choice B is incorrect. The y-axis of this graph doesn't depict baseline levels of REM sleep, but rather shows REM sleep as a percent of baseline. Choice D is incorrect. The graph doesn't depict REM sleep while in the water for the seals in the study. Additionally, we're told fur seals get no REM sleep while in the water, which is significantly different to the values shown in the graph for after they return to land.

- C. The bite force estimates produced by Bates and Falkingham and by Cost et al. were similar to each other, while the estimates produced by Meers and by Gignac and Erickson each differed substantially from any other estimate.

Choice C is the best answer because it accurately describes data from the table that support Barrett and Rayfield's suggestion about bite force estimates. According to the text, Barrett and Rayfield believe that estimates of dinosaur bite force may be strongly influenced by the methods used to produce them—that is, that different methods may produce significantly different results. The table shows that the studies by Bates and Falkingham and by Cost et al. used the same estimation method (muscular and skeletal modeling) and produced similar bite force estimates (approximately 35,000–57,000 newtons and 35,000–63,000 newtons, respectively). The study by Meers, however, used body-mass scaling and produced a much higher bite force estimate (183,000–235,000 newtons), while the study by Gignac and Erickson used tooth-bone interaction analysis and produced a much lower bite force estimate (8,000–34,000 newtons). The fact that one method produced similar estimates in two different studies and that two different methods used in other studies produced substantially different estimates supports the idea that dinosaur bite force estimates are significantly influenced by the methodology used to produce them.

Choice A is incorrect because it inaccurately describes data from the table. The table does show that the studies by Meers and by Cost et al. used different estimation methods and produced very different ranges of estimated dinosaur bite force, which would support Barrett and Rayfield's suggestion that different methodologies may produce significantly different estimates. However, the table doesn't show that the study by Meers produced the lowest estimated maximum bite force while the study by Cost et al. produced the highest. In fact, the study by Meers estimated a maximum bite force of approximately 235,000 newtons, which is the highest of all the estimated maximums. Choice B is incorrect. Although the data from Gignac and Ericson's study are accurately described, a single set of findings from one study using only one methodology can't show that different methodologies may produce significantly different dinosaur bite force estimates, as Barrett and Rayfield suggest. Choice D is incorrect. Although the table shows that the maximum bite force estimated by Cost et al. was higher than that estimated by Bates and Falkingham, the difference is relatively small; in fact, both teams estimated a minimum bite force of approximately 35,000 newtons and a maximum bite force close to approximately 60,000 newtons. Because these findings demonstrate that a single methodology (muscular and skeletal modeling) produced similar overall results in two studies,

the findings don't support Barrett and Rayfield's suggestion that different methodologies may produce significantly different dinosaur bite force estimates.

Q10

A

- A. When plants exposed to air from whitefly-free plants were infested, they produced more jasmonic acid than did control plants, whereas when plants exposed to air from whitefly-infested plants were infested, they produced less jasmonic acid and more salicylic acid than did control plants.

Choice A is the best answer because it best describes data from the graph that support the researchers' conclusion about whitefly-induced defenses in tomato plants. According to the text, tomato plants produce airborne chemicals that prompt nearby tomato plants to increase their production of appropriate defensive chemicals, such as jasmonic acid in response to herbivory and salicylic acid in response to microbial infection. The text goes on to indicate that researchers concluded that whiteflies induce tomato plants to emit chemicals that cause nearby plants to defend against microbial infection instead of herbivory from whiteflies. The graph shows that control tomato plants (plants that weren't exposed to any chemicals from nearby tomato plants and weren't subsequently infested with whiteflies) produced about 140 nanograms of jasmonic acid per gram of dry weight and about 120 nanograms of salicylic acid per gram of dry weight, which indicates the baseline levels of these defensive chemicals in tomato plants. The graph also shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight when subsequently infested by whiteflies, which is more than the amount produced by the control plants. This suggests that when infested with whiteflies, tomato plants that weren't previously exposed to air from nearby whitefly-infested plants begin producing higher levels of jasmonic acid to defend against the infestation. Finally, the graph also shows that plants exposed to air from whitefly-infested plants and then subsequently infested by whiteflies produced approximately 100 nanograms of jasmonic acid per gram of dry weight (which is less than control plants did) and approximately 175 nanograms of salicylic acid per gram of dry weight (which is more than control plants did). This suggests that whiteflies cause tomato plants to emit chemical signals that decrease nearby plants' defenses against whitefly herbivory relative to baseline levels and redirect their defenses toward microbial infection, thereby supporting the researchers' conclusion that whiteflies cause tomato plants to emit airborne chemical signals that induce nearby plants to preferentially defend against microbes even when experiencing herbivory from whiteflies.

Choice B is incorrect. Although the graph shows that tomato plants exposed to whitefly-infested plants produced less jasmonic acid than salicylic acid (about 100 nanograms of jasmonic acid per gram of dry weight vs. about 175 nanograms of salicylic acid per gram of dry weight), and although this might suggest that those plants are therefore better defended against microbial attack than against herbivory, this wouldn't be sufficient to conclude that this is the case:

without reference to the relative levels of defensive chemicals produced by the plants in the other experimental conditions, it would be impossible to conclude that exposure to air from whitefly-infested plants either induces or diminishes these plants' defenses to different kinds of threats. Furthermore, the graph shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight and about 130 nanograms of salicylic acid per gram of dry weight, not that they produced the same amount of the two acids. Choice C is incorrect because it inaccurately describes the data in the graph. The graph shows that plants in all three conditions produced both jasmonic and salicylic acid, not that plants exposed to air from whitefly-infested plants produce no jasmonic acid at all. Choice D is incorrect. Although the graph shows that plants exposed to air from whitefly-infested plants produced about 100 nanograms of jasmonic acid per gram of dry weight when infested, which is less than the amount produced by control plants (about 140 nanograms per gram of dry weight), and although the graph shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight when infested (which is more than the amount produced by control plants) and about 130 nanograms of salicylic acid per gram of dry weight (which is more than the approximately 120 nanograms per gram of dry weight produced by control plants), this wouldn't support the researchers' conclusion that plants exposed to air from whitefly-infested plants preferentially defend against microbial infection—that is, that they increase their levels of salicylic acid and decrease their levels of jasmonic acid relative to control plants. Rather, the data would suggest that exposure to air from whitefly-infested plants reduces plants' defenses against herbivory, but these data wouldn't indicate that whiteflies induce plants to redirect their defenses to guard against microbial infection. Furthermore, the data concerning plants exposed to air from whitefly-free plants merely indicate that these plants increase their defenses against both herbivory and microbial infection when infested by whiteflies.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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